TinyML - Driven Occupancy and Appliance Control System for Classroom Energy Reduction

Jayshri
Computer Science in AI/ML
Sai Vidya Instuite of Technology
Bengaluru, India
jayshri.j@saividya.ac.in

Arun Arya
Computer Science in AI/ML
Sai Vidya Instuite of Technology
Bengaluru, India
arunarya.23aiml@saividya.ac.in

Balaji M
Computer Science in AI/ML
Sai Vidya Instuite of Technology
Bengaluru, India
balajim.23aiml@saividya.ac.in

Chandan Hegde
Computer Science in AI/ML
Sai Vidya Instuite of Technology
Bengaluru, India
chandanhegde.23aiml@saividya.ac.n

Manjunatha S
Computer Science in AI/ML
Sai Vidya Instuite of Technology
Bengaluru, India
manjunathas.23aiml@saividya.ac.in

***Abstract*—A TinyML-based Smart Classroom Energy Optimization System is proposed to reduce unnecessary energy consumption caused by lights and fans remaining ON in unoccupied classrooms. PIR, LDR, and temperature sensors were used solely for data collection, capturing motion, ambient light, and environmental conditions. The collected dataset trained a lightweight Feedforward Neural Network optimized for TinyML. Instead of deploying the model on physical hardware, a hybrid approach integrates the trained TinyML model into a software dashboard, enabling real-time occupancy prediction and simulated energy-control actions. An interactive 3D classroom simulation further visualizes sensor behavior, occupancy, and automated decisions. This hybrid system offers a low-cost, scalable, and highly demonstrable prototype for smart classroom energy management.**

Keywords*—*TinyML, Occupancy Prediction, Smart Classroom, Energy Optimization, 3D Simulation

I. INTRODUCTION

Educational institutions often waste electricity as lights and fans remain ON in empty classrooms. Traditional systems such as manual switches and basic motion sensors lack contextual understanding and predictive control, resulting in continued inefficiency. To address this challenge, intelligent energy-management solutions are needed that leverage data-driven decision-making. This project introduces a hybrid TinyML-based system that trains an occupancy-prediction model using real environmental data collected through sensors at home. The trained model is deployed virtually through a dashboard and complemented with a 3D classroom simulation, enabling an intuitive and cost-effective demonstration of smart energy optimization.

II. PROBLEM STATEMENT

Educational institutions experience substantial energy wastage because fans and lights are frequently left on in unoccupied classrooms. Current solutions depend on manual switching or simple motion sensors, which often fail due to their lack of accuracy, contextual understanding, and predictive capability.

These limitations result in continuous power loss, higher operational expenses, and reduced sustainability on campuses. Therefore, there is a clear need for an intelligent, data-driven approach that can accurately understand real-time classroom usage patterns and automatically optimize energy consumption without relying on complex or expensive infrastructure

III. OBJECTIVES

OBJECTIVE 1: To develop an intelligent occupancy-prediction model trained using real environmental data collected from controlled home-based experiments, enabling accurate identification of occupancy patterns suitable for smart classroom energy optimization.

OBJECTIVE 2: To design a hybrid, software-based energy optimization system that deploys the TinyML model within an interactive dashboard for real-time decision-making without hardware dependency.

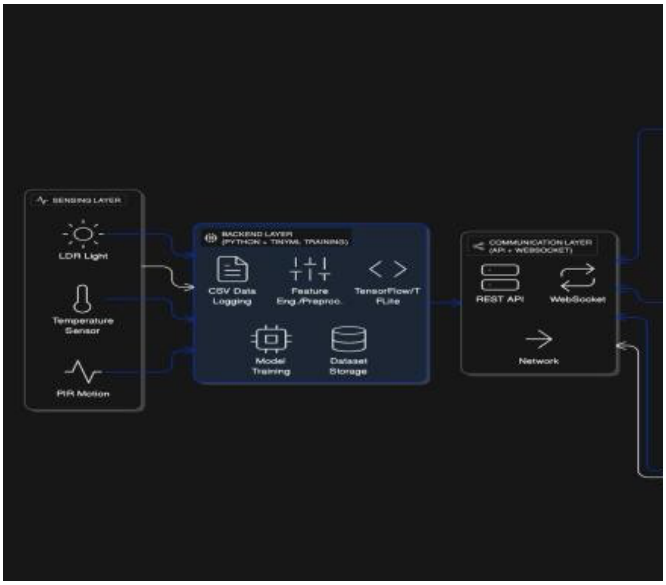
OBJECTIVE 3: To create an immersive 3D classroom visualization that dynamically showcases occupancy predictions and automated energy-control actions, providing a clear and engaging virtual representation of smart classroom behavior.

VI. IMPLEMENTATION

The system was implemented through a hybrid workflow combining sensor-based data collection, machine learning model development, and software-based visualization. PIR, LDR, and temperature sensors were used to collect environmental data in controlled conditions, forming the dataset for training the occupancy-prediction model. To ensure data quality, multiple readings were recorded under varied lighting, movement, and temperature scenarios to accurately represent real classroom conditions.

V. METHODOLOGY / SYSTEM ARCHITECTURE

TinyML-based Feedforward Neural Network modeling, real-time data acquisition, virtual dashboard deployment, and 3D classroom simulation for predictive energy optimization.



Python (data processing and model training), TensorFlow/TinyML (model optimization), HTML–CSS–JavaScript (Web dashboard), Three.js (3D simulation), and Visual Studio Code & Arduino IDE for development—entirely built using opensource technologies.

The methodology of the proposed system is structured into four major phases: sensor-based data collection, machine learning model development, hybrid TinyML deployment, and virtual classroom simulation. Each phase is designed to ensure that the system accurately captures environmental behavior, learns occupancy patterns, and demonstrates intelligent energy optimization within a fully software-driven environment.

The process begins with collecting environmental data using PIR, LDR, and temperature sensors positioned in a controlled indoor setting. These sensors individually capture motion activity, ambient light intensity, and temperature variations—factors known to correlate strongly with room occupancy. The ESP32 microcontroller streams raw sensor values to a laptop via serial communication. Each reading is timestamped and manually labeled as either occupied or empty based on observed ground truth. This labeled dataset undergoes preprocessing, including noise filtering, interpolation of missing values, normalization of sensor ranges, and alignment of timestamps to form a consistent, structured dataset suitable for machine learning.

Following dataset preparation, a lightweight Feedforward Neural Network (FNN) is developed using Python and TensorFlow. The architecture is deliberately kept shallow—typically two hidden layers—to ensure compatibility with TinyML constraints while retaining sufficient expressive power for occupancy classification. Hyperparameters such as learning rate, batch size, and activation functions are optimized through iterative experimentation. During training, evaluation metrics including accuracy, F1-score, confusion matrix, and validation loss are monitored to measure the model's learning progression and prevent overfitting.

Once the model achieves stable performance, the network is quantized and converted into TensorFlow Lite Micro format, reducing computational footprint and enabling near real-time inference.

methodology that demonstrates smart classroom energy optimization in a completely software-driven, accessible, and hardware-light environment, while preserving the core principles of TinyML-based edge intelligence.

The hybrid deployment phase shifts execution away from embedded hardware and into a virtual runtime environment implemented through a web-based dashboard. Here, the TinyML model is loaded in a lightweight inference engine built using JavaScript or Python backend services. The dashboard accepts live sensor inputs or simulated streams, processes them through the TinyML model, and instantly displays occupancy predictions. Based on these predictions, the system generates automated energy-control decisions (e.g., "Lights OFF when Empty," "Fan ON when Occupied"), which are showcased through UI indicators. This hybrid approach maintains the logic and workflow of an actual TinyML deployment while removing the complexity of microcontroller integration.

To further enhance interpretability and engagement, a full 3D classroom simulation is constructed using modern web visualization technologies such as Three.js or Babylon.js. This simulation renders a virtual classroom environment where lighting brightness, fan rotation, and occupancy markers change dynamically according to model predictions. Sensor behavior is visually represented through interactive graphs or icons showing motion detection, light intensity, and temperature fluctuations. This visual layer bridges the gap between raw data and system behavior, allowing users, evaluators, and stakeholders to clearly understand how the TinyML model drives energy-management decisions.

Together, these phases form a cohesive and scalable methodology that demonstrates smart classroom energy optimization in a completely software-driven, accessible, and hardware-light environment, while preserving the core principles of TinyML-based edge intelligence

A Feedforward Neural Network was developed using Python and TensorFlow, optimized for TinyML deployment. The model underwent preprocessing steps such as normalization, feature selection, and dataset balancing to improve accuracy and reduce overfitting. Although it was designed to be deployable on microcontrollers, the trained model was instead integrated into a web-based dashboard, where it performs real-time inference and simulates automated ON/OFF decisions for lights and fans. The dashboard offers live input sliders representing sensor values, enabling users to test predictions without hardware.

FNN vs Classical Machine Learning Models – Performance Comparison

To evaluate the suitability of the Feedforward Neural Network (FNN) for TinyML deployment, its performance was compared against three commonly used classical machine-learning models: XGBoost, Gradient Boosting, and K-Nearest Neighbors (KNN). The comparison included accuracy, precision, recall, F1-score, inference speed, ROC-AUC, and deploy ability on edge devices.

Across all metrics, the FNN performed competitively with traditional ML models, achieving 98.7% accuracy with only 833 parameters, making it the most lightweight and edge-deployable option. While XGBoost offered slightly higher accuracy (~99%), its model size and computational load make it unsuitable for microcontroller-level deployment. Gradient Boosting and KNN also performed well but required higher memory and compute, making them more appropriate for server-based inference rather than TinyML devices.

Inference-speed analysis showed that the FNN achieved 0.0482 ms per prediction, significantly faster and more stable than Gradient Boosting, and only slightly slower than KNN. The ROC curves of all models remained above 0.995, indicating excellent separability. Confusion-matrix comparisons further confirmed the FNN's strong generalization capability across both occupied and unoccupied classes.

This analysis validates that the FNN provides the best balance of accuracy, model size, inference speed, and edge compatibility, making it ideal for TinyML-based real-time classroom occupancy prediction.

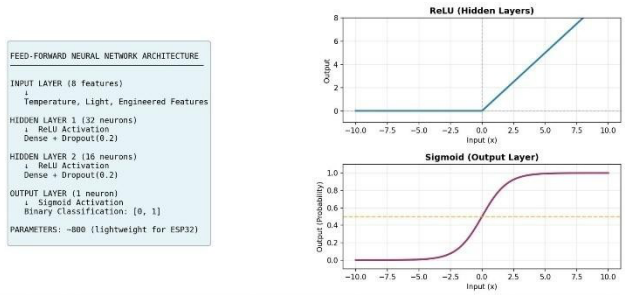
IV . LITERATURE REVIEW

Recent studies have explored intelligent energy management using sensor data, machine learning, and embedded systems. Research on occupancy detection frequently uses PIR sensors, LDR sensors, ultrasonic sensors, Wi-Fi signals, CO_2 levels, and thermal data to determine human presence, but most systems rely on traditional machine learning or rule-based logic, limiting accuracy and adaptability. Many works highlight that PIR sensors alone often generate false triggers due to limited detection range and sensitivity to micro-movements, while camera-based systems raise privacy concerns and require high computational power.

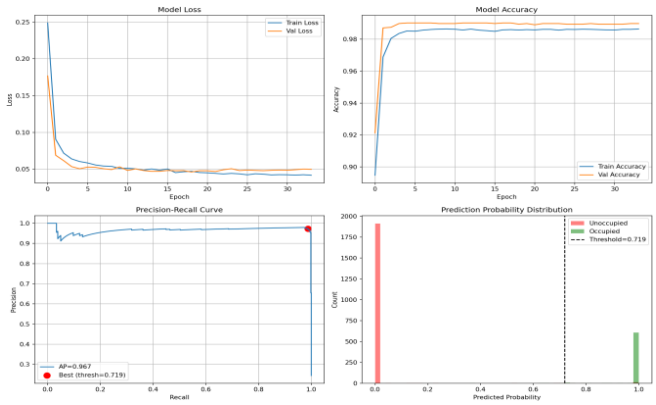
In recent years, TinyML-based approaches have gained attention for enabling on-device inference with low power consumption, especially in smart-building and smart-campus applications. Studies show that TinyML models can run efficiently on microcontrollers for tasks such as occupancy classification, appliance optimization, and anomaly detection. However, existing solutions often require full hardware deployment, lack advanced visualization tools, or depend heavily on cloud connectivity for analytics and monitoring. Some research emphasizes digital twins for energy simulation, but these systems are usually expensive and complex, restricting their use in educational environments.

VII. RESULTS

The proposed system achieved strong predictive performance, with the TinyML-optimized neural network recording an accuracy of 98% in identifying occupancy from environmental sensor data. The web dashboard successfully displayed real-time predictions, simulated appliance-control decisions, and presented clear visualization of model outputs. The 3D classroom environment further demonstrated the system's behavior by animating sensor states, occupancy changes, and corresponding energy-saving actions. Together, these results validate the effectiveness of the hybrid approach and show that intelligent energy optimization can be demonstrated without requiring physical relay-based hardware.

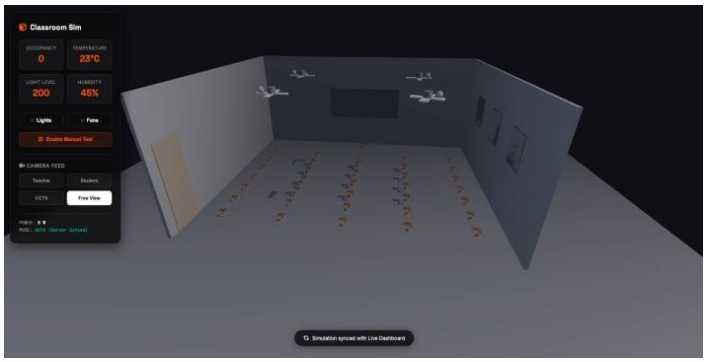


Property	ReLU	Sigmoid
Formula	max(0, x)	1/(1 + e^(-x))
Output Range	[0, ∞)	[0, 1]
Used In	Hidden Layers	Output Layer
Purpose	Learn Complex Patterns	Probability Output
Gradient	Constant (1) for x>0	Vanishes at extremes



Model	Accuracy	Precision	Recall	F1-Score	ROC-AUC	Inference Speed (ms)	Parameters	Deployment Suitability
FNN (Proposed)	0.9871	0.9707	0.9946	0.9824	0.995	0.0482 ms	833	EDGE (TinyML)
XGBoost	0.9934	0.9846	0.9999	0.9919	0.999	0.0461 ms	~10K+	Cloud/Server Only
Gradient Boosting	0.9918	0.9811	0.9971	0.9870	0.997	0.5076 ms	High	Cloud/Server Only
K-Nearest Neighbors	0.9903	0.9775	0.9963	0.9860	0.996	0.0146 ms	Very High	Cloud/Server Only

VIII. CONCLUSION & FUTURE WORK



Conclusion: The project successfully developed a hybrid TinyML-based system capable of predicting occupancy with 98% accuracy using simple environmental sensors. The integration of a real-time dashboard and a 3D classroom simulation enabled an effective, low-cost demonstration of intelligent energy optimization without requiring physical hardware deployment. This hybrid framework proves that smart energy management can be achieved through data-driven modeling and virtual visualization.

Future Work: Future enhancements may include deploying the TinyML model on embedded hardware for real-time on-device inference, integrating additional environmental inputs, expanding the 3D environment for multi-room simulations, and incorporating actual relay-based energy control for full automation.

IX. REFERENCES

I. **V. A. Pawar, M. K. Rathod**, and **M. S. Anant**, "Integrating Analog PIR Sensor Telemetry with TinyML Inference," in *Advances in Signal Processing and Intelligent Recognition Systems*, Springer, pp. 401–410, 2023.

II. **J. Lin, S. Han, et al.**, "Tiny Machine Learning: Progress and Futures," arXiv preprint arXiv:2403.19076, 2024.

III. **R. Garcia, M. Ahmad**, and **K. Singh,** "PIR Sensor-Based Occupancy Monitoring in Smart Buildings: A Review of Methodologies and Machine Learning Approaches," *Sensors*, vol. 24, no. 5, p. 1533, 2024.

IV. A Review of Methodologies and Machine Learning Approaches," *Sensors*, vol. 24, no. 5, p. 1533, 2024.

V. **M. Jurado, P. Stanley,** and **T. Li,** "Benchmarking Energy and Latency in TinyML: A Novel Method for Resource-Constrained AI," arXiv preprint arXiv: 2505.15622, 2025.

VI. **A. S. Afolabi, O. A. Akinola, O. A. Odetoye & E. Adetiba**, "Indoor Occupancy Detection Using Machine Learning and Environmental Sensors," *Journal of Smart Science and Technology*, vol. 5, no. 1, pp. 17–39, 2025.

• Occupancy detection using environmental sensors and ML.

VII. **J. Chen et al.**, "A Multi-Modal Data-Driven Meeting Room Occupancy Detection for Building Energy and Space Optimization," *Building and Environment*, 2025.

• Multimodal sensor fusion for occupancy and energy optimization.

VIII. **H. Elkhoukhi**, "A Review of Occupancy Sensing Technologies and Approaches in Smart Buildings," 2025.

• Survey of occupancy sensing technologies and ML approaches.

IX. **W. Song et al.**, "Impact of Occupancy Behavior on Building Energy Efficiency: A Review," 2025.

• Shows how occupancy behavior influences energy use.

X. **M. A. Abuhussain, B. S. Alotaibi, Y. A. Dodo, A. Maghrabi & M. S. Aliero**, "Multimodal Framework for Smart Building Occupancy Detection," *Sustainability*, vol. 16, no. 10, 2024.

• Multimodal smart-building occupancy detection framework.